STYLEINSTRUCTCAPS: A Benchmark for Controllable Speaking-Style Captioning in Audio-Language Models

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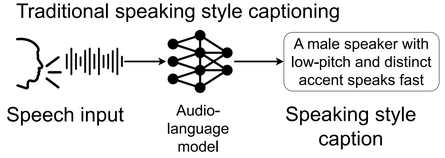
Project page | Paper | Shreeram's homepage

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AACL-IJCNLP 2026 Hengqin | China

WHAT IS SPEAKING-STYLE CAPTIONING?

Describe in natural language **how** speech is delivered, not *what* is said.



Vocal timbre	Prosody	Expressivity
voice quality	pitch, rate, rhythm	emotion, emphasis

NO UNIFIED FORMULATION

UH-OH!

Each research community describes a **different aspect** of style:

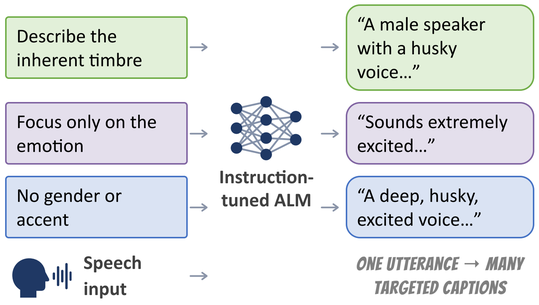
- **Speech generation** — Speaker attributes: "A male speaker with an American accent has a husky voice…"
- **Emotion understanding** — Emotion: "The speaker sounds extremely excited, with a rising pitch at the end…"

A single caption mixes all of these, so systems are hard to compare.

STYLEINSTRUCTCAPS

SSC as **instruction following**: the instruction picks *which* aspect of style to describe.

$$\hat{y} \sim p_\theta(y \mid x, s)$$ **x** speech **s** instruction $\hat{y}$ caption

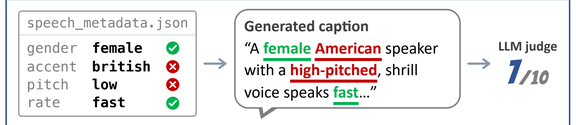


FIVE AXES OF EVALUATION

Axes 1–4 use open-weight LLM judges (a three-judge panel for groundedness and hallucination, Qwen3-32B for instruction following); axis 5 uses caption embeddings.

1 METADATA GROUNDEDNESS

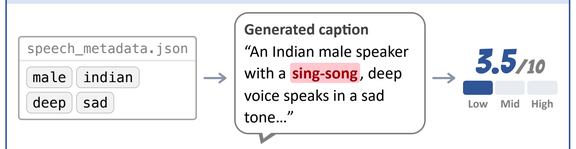
Is the caption faithful to the metadata? — Meta score 1–10 ↑



The judge checks the caption against **ground-truth metadata**.

2 HALLUCINATION

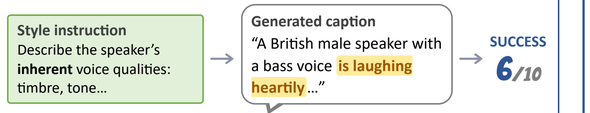
Does it invent unsupported attributes? — Hall. 0–10 ↓ + severity



"Sing-song" is **not supported** by the metadata.

3 INSTRUCTION FOLLOWING

Does it describe only what was asked? — IFA score 1–10 ↑ IFA rate % ↑



Inherent traits are correct, but a **situational emotion** slips in.

4 GENERALIZATION

Does it transfer to unseen data? — Meta / Hall. IFA score

Unseen speech data

GigaSpeech: 600 YouTube and podcast clips

36 unseen instructions

Narrative	Describe the emotional…
Key-value	{valence: 1–7, arousal…
Tags / list	Ranked list of up to t…
Minimal	One sentence, no pitch…
Persona	As a call-center QA ev…

Training prompts are mostly **narrative**.

5 SPEAKER-STYLE CONSISTENCY

Are speaking-style captions consistent within a speaker? — EER ↓ AUC ↑ ARI · NMI ↑

Speaker-caption verification



Speaker-caption clustering



k-means on caption embeddings, k = # speakers

In-domain and on **unseen VCTK speakers**.

STYLEINSTRUCTCAPS-DB

2.7K	1.03M	6
hours of speech	training utterances	parallel captions per utterance

- Built on **ParaSpeechCaps** (Expresso, EARS, VoxCeleb, Emilia)
- Captions generated from rich per-utterance metadata
- Train/val captions: **Mistral-Small-24B**; eval captions: a different LLM, **Llama-3.1-8B**

Example: EARS speaker p047, "amazement" free-form

`female` `american` `high-pitched` `measured speed` `awed` `crisp` `nasal` `authoritative` `loud`

"Oh, really? You were so good at that. I love to just sit here and listen to you play…"

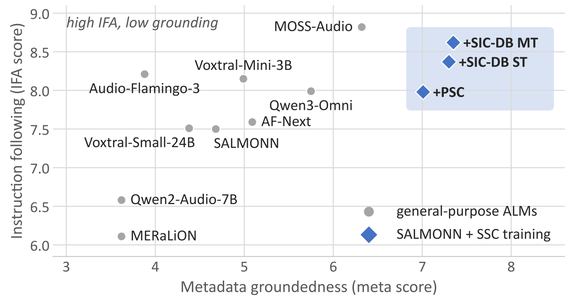
Speaker idiosyncratic	"A high-pitched, crisp, and nasal voice with an American accent, delivered by a female speaker…"
Situational / contextual	"Speaking with measured speed and authority… cuts through a slightly clean environment…"
Expressive / emotional	"A voice filled with awe and admiration, conveyed through a rising pitch contour…"
Linguistic / pragmatic	"A steady, measured rhythm, punctuated by deliberate pauses that emphasize key phrases…"
Perceptual / listener-centric	"Listening to this voice is like being drawn into a conversation with a charismatic individual…"
Overall style summary	"Imagine a female speaker with a high-pitched, crisp, nasal voice… expressing awe…"

KEY RESULTS

WHOA!

SALMONN tuned on PSC or SIC-DB (ST/MT: single/multi-task).

FOLLOWING INSTRUCTIONS ≠ GROUNDED UNDERSTANDING



85.9% HUMAN AGREEMENT — **Are LLM judges reliable?** They agree with human listeners on 85.9% of caption pairs (κ = 0.72).

WHAT WE FOUND

BOOM!

Following instructions does **not** mean understanding style. **Instruction-diverse SSC training** improves grounding, control and speaker consistency.

1.35 best meta score (MT) — **SSC TRAINING: BIGGEST GROUNDING GAINS**
Every SSC-tuned SALMONN beats the best general model (6.32).

8.21 AF-3 IFA score — **FOLLOWING INSTRUCTIONS ≠ GROUNDING**
AF-3 follows instructions well but is weakly grounded (3.88).

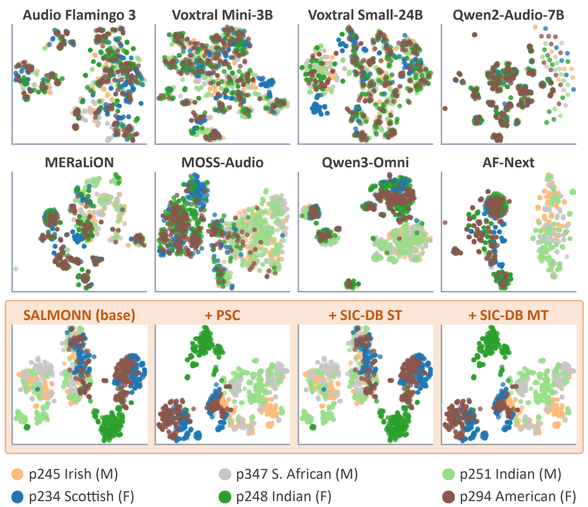
$\rho \approx 0$ grounding vs. hallucination — **LOW HALLUCINATION ≠ FAITHFUL**
MERaLiON hallucinates least, yet is among the least grounded.

75% IFA rate with MT (base 58.5%) — **DIVERSE INSTRUCTIONS UNLOCK CONTROL**
SSC data alone does not help (52.9%); instruction diversity does.

3.34 AF-3 IFA on key-value prompts (vs. 7.26) — **GENERALIZATION IS STILL OPEN**
New prompt formats break some models; unseen audio sees no gain.

SPEAKER CLUSTERS IN CAPTION SPACE

t-SNE of caption embeddings, six unseen VCTK speakers.



Task-specific instruction tuning improves speaker-style consistency.